

Chapter 2

DNA viruses

DNA viruses

Objectives

- Explain the properties of DNA virus
- Describe the pathogenesis of DNA virus
- Explain the clinical features of diseases caused by DNA viruses
- Describe the diagnosis of diseases caused by DNA viruses

Adenoviruses

- Family: Adenoviridae, Two Genera
 - Aviadenovirus – Infects birds
 - Mastadenovirus – Infects mammals
- Linear, ds DNA, Non-enveloped
- More than 49 human stereotypes known
- Common stereotypes: 1 -8, 11, 21, 35, 37, 40 & 41
 - ▶ Types 40 & 41 are enteric pathogens

Not easily affected by:

- External environment, Low PH, Bile salts & proteolytic enzymes
- Can replicate to high titer in the gastro intestinal tract

➤ **Infections:** More than 100 serotypes of which, At least **57** (HAdV-1 to 57) infect humans (seven species (Human adenovirus A to G))

- ✓ Respiratory mainly species of HAdV-B and C
- ✓ Eye (HAdV-B and D)
- ✓ Genito- urinary tract (HAdV-F types 40, 41, HAdV-G type 52)

PATHOGENESIS

- Transmitted by **respiratory** or **feco-oral** contact.
- Show **three** types of interactions with the infected cells.

➤ **lytic infection**

- infect **mucoepithelial cells** in the respiratory tract, gastrointestinal tract, and conjunctiva or cornea, causing damage of these cells directly
- Enter via epithelium, replicate and spread to lymphoid tissue

➤ **latent infection** (Group B and C)

- **lymphoid** and **adenoids**, tonsils, and Payer's patches
- Reactivated in patients infected with other agents

➤ **Oncogenic transformation**

- from groups A and B have the property for oncogenic transformation in rodent cells
- Uncontrolled cell growth and replication occur

Host immunity

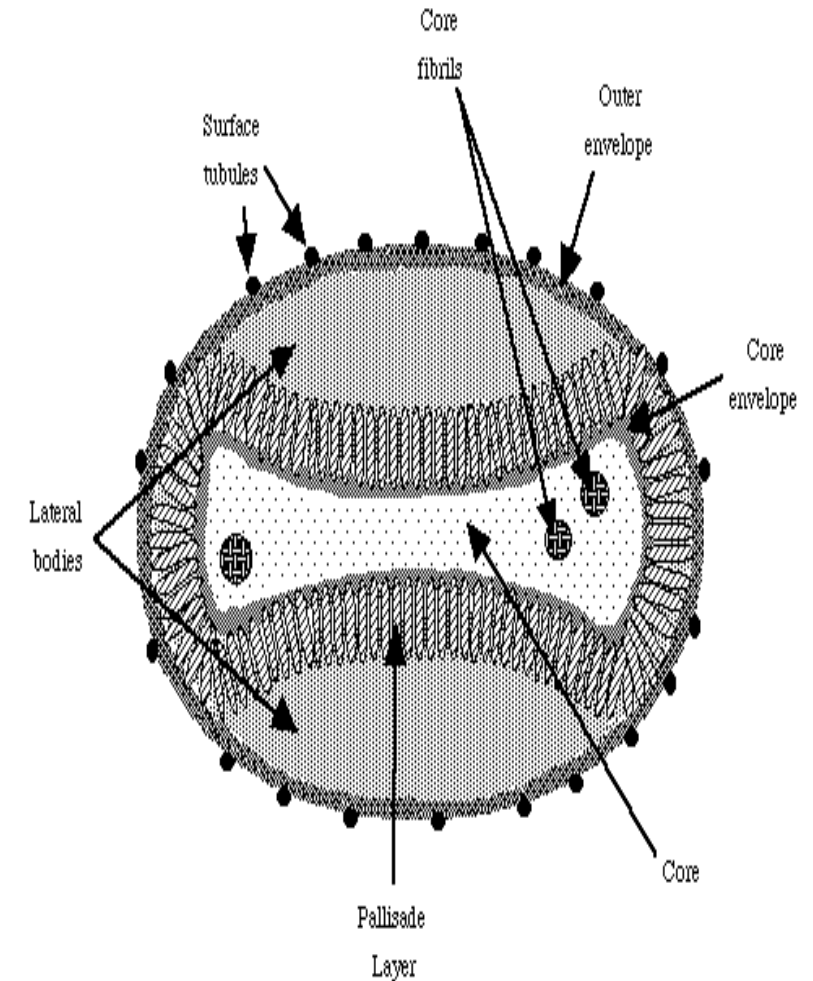
- Adenovirus infection is characterized by development of an effective and long-lasting immunity against reinfection
- Both cell-mediated immunity (CMI) and humoral immunity are important.
 - CMI plays an important role in **limiting proliferation** and **outgrowth** of adenoviruses
 - different disease is seen in persons whose CMI is compromised.
- The **antibodies** protect the person from **reinfection** with the **same serotype** but not other serotypes

Laboratory Diagnosis

- Specimens depend on the nature of the clinical illness
 - throat swab, nasopharyngeal aspirate, bronchial lavage, conjunctival swab, corneal scraping, urine, and feces
- **Microscopy**
 - Certain serotypes of enteric adenoviruses are seen in stool specimens by EM
 - Enteric serotypes are non-cultivable (40,41 and 52)
- **Direct antigen detection**
 - ELISA or direct fluorescent antibody test
- **Isolation of the virus**
 - Cell culture establishes the specific diagnosis
 - Isolated on primary human embryonic kidney cells, HeLa, Hep-2, and KB cells

POXVIRUSES

- The largest and most complex of viruses infecting humans.
- Enveloped virus, complex structure
- ds DNA, linear, brick shaped
- Replicate in cytoplasm
- DNA 3 %,protein 90 %, lipid 5 %
- Infection characterized-skin rash
- Has large number of viruses
- Similar morphologically, share a common nucleocapsid antigen
- Relatively resistant
- Chemicals, heat, cold, drying



POXVIRUSES.....

➤ Two subfamilies based on whether they infect **vertebrate** or **insect** hosts.

- **Chordopoxvirinae (vertebrates):** six genus

- **Orthopoxvirus, vaccinia virus (smallpox)**

- **variola virus** (VV) and **vaccinia** are the best known

- VV strains are divided into variola major (25-30% fatalities) and variola minor (same symptoms but less than 1% death rate)

- Overwhelming toxemia was the usual cause of death in patients with smallpox

- **Parapoxvirus, Orf virus**

- **Entomopoxvirinae (insect)**

Family = Poxviridae

- Subfamily
Genus
Members

1 Avipoxvirus
virus

Fowlpox

2 Capripoxvirus
Sheeppox virus

3 Leporipoxvirus

Myxoma virus

1 Chordopoxvirinae
(vertebrates)

4 Molluscipoxvirus

Molluscum contagiosum

5 Orthopoxvirus
vaccinia virus (smallpox)

6 Parapoxvirus
virus

Orf

7 Suipoxvirus

Swine pox

Pathogenesis of poxvirus

- Portal of entry → mm of URT ↓

Multiplication in lymphoid tissue local LN ↓

Transient viremia RES ↓

Secondary severe viremia ↓

Clinical disease ↓

Lesions in mouth and URT ↓

Discharge in environment

Epidermis → skin pustules → scabs (virus)

- Histopathologic examination of the skin showed proliferation of the prickle-cell layer.
 - cytoplasmic inclusions

Smallpox



Immunity

- All viruses within the *Orthopoxvirus* genus are so **closely related antigenically**.
- Infection with **one induces** an immune response that reacts with **all** other members of the group.
- An attack of smallpox gave complete protection against reinfection.
- Both CMI and adaptive immunity are involved
- Cell-mediated immunity is probably more important than circulating antibody.
- Patients who have defects in both cellular immune response and antibody response develop a progressive, usually fatal disease upon vaccination.

DIAGNOSIS

- **Specimens**

- Skin lesions, such as vesicular fluid, were the specimens of choice

- **Direct EM**

- Guarnieri bodies, the characteristic eosinophilic inclusion bodies

- **Isolation and Identification of virus on Cell cultures (chick embryo)**

- Vesicular fluid → CAM parapoxviruses, molluscum contagiosum and tanapox virus do not grow on CAM
 - Orthopoxviruses grow well in cultured cells

- **Serology**

- **PCR**

Human Herpesviruses

- '*Herpein*' - 'to creep' = chronic/latent/recurrent infections

Herpes virus types that infect humans

- Herpes simplex I & II: (cold sores, genital herpes)
- Varicella zoster : (chicken pox, shingles)
- Cytomegalovirus (microcephaly, infectious mono)
- Epstein-Barr virus (mononucleosis, Burkitt's lymphoma)
- Human herpesvirus 6 & 7 (Roseola)
- Human herpesvirus 8 (Kaposi's sarcoma)

Herpes virus Virion

- Family **Herpesviridae**, Icosahedral, Enveloped, large and linear ds DNA
- More than 35 proteins in virion
- Envelope: 8nm spikes, viral glycoproteins, Fc receptors.
- Replication nuclear, bud from nuclear membrane
- Infection: Lytic, latent and recurrent
- Herpes viruses encode several glycoproteins for viral attachment, fusion, and for escaping immune control.
- As enveloped viruses, the herpes viruses are sensitive to
 - fat solvents, such as alcohol, chloroform, ether, and bile salts

Virion has 4 basic structures

1. Envelope: associated glycoproteins-spikes
2. Tegument: Protein-filled region between capsid and envelope
 - viral proteins and enzymes that help initiate replication.
3. Icosahedral capsid-162 capsomers
4. DNA-containing core: Toroidal (DNA around protein),

Properties Distinguishing the Herpesviruses

Subfamily	Virus	Primary Target Cell	Site of Latency	Means of Spread
Alphaherpesvirinae				
Human herpesvirus 1	Herpes simplex type 1	Mucoepithelial cells	Neuron	Close contact (sexually transmitted disease)
Human herpesvirus 2	Herpes simplex type 2	Mucoepithelial cells	Neuron	
Human herpesvirus 3	Varicella-zoster virus	Mucoepithelial and T cells	Neuron	Respiratory and close contact
Gammapherpesvirinae				
Human herpesvirus 4	Epstein-Barr virus	B cells and epithelial cells	B cell	Saliva (kissing disease)
Human herpesvirus 8	Kaposi sarcoma-related virus	Lymphocyte and other cells	B cell	Close contact (sexual), saliva?
Betaherpesvirinae				
Human herpesvirus 5	Cytomegalovirus	Monocyte, granulocyte, lymphocyte, and epithelial cells	Monocyte, myeloid stem cell, and ?	Close contact, transfusions, tissue transplant, and congenital
Human herpesvirus 6	Herpes lymphotropic virus	Lymphocytes and ?	T cells and ?	Saliva
Human herpesvirus 7	Human herpesvirus 7	Like human herpesvirus 6	T cells and ?	Saliva

Herpes Simplex Virus

- Extremely host-adapted viruses that can cause a wide variety of illness
- There are two types of the HSVs:
 - herpes simplex virus type 1 (HSV-1)
 - herpes simplex virus type 2 (HSV-2).
- Both types are closely related in their DNA homology, antigenic determinants, tissue tropism and disease symptoms
- Differ in epidemiology
 - HSV-1 is transmitted primarily by contact with infected saliva
 - HSV-2 is transmitted by sexual contact or by genital tract infection to newborn from an infected mother.

Herpes Simplex Viruses (1 & 2)

- They grow rapidly and are highly cytolytic.
- Responsible for a spectrum of diseases, ranging from
 - Gingivostomatitis to keratoconjunctivitis,
 - Encephalitis,
 - Genital disease, and
 - Infections of newborns.
- The herpes simplex viruses establish latent infections in **nerve cells**; recurrences are common
- Characteristic histopathologic changes include
 - Ballooning of infected cells
 - Cowdry type A intranuclear inclusion bodies
 - Formation of multinucleated giant cells.
- Cell fusion provides an efficient method for cell-to-cell spread of HSV, even in the presence of **neutralizing antibody**.

Pathogenesis

- HSV causes disease by direct cytopathologic effects and shows three unique biological properties
- **Neurovirulence** : invade and multiply in the nervous system
- **Latency**: Establishment and maintenance of latent infection in the nerve cell ganglion
 - HSV-1 affects most commonly the trigeminal ganglion
 - HSV-2 affects the sacral nerve root ganglion (S2–S5)
- **Reactivation**: Reactivation and replication of latent HSV
 - Can lead to overt or covert recurrent infections and excretion of HSV
 - HSV-1 reactivates more frequently in the oral cavity
 - HSV-2 reactivates eight to ten times more frequently in the genital parts

Reactivation

- Stress Physical/ psychological
- Pneumococcal infection
- Meningococcal infection
- Fever
- Irradiation
- Menstruations
- Immunosuppression
- Cytotoxic drugs
- HIV
- Organ graft etc

Clinical Syndromes

- Causes a wide variety of clinical manifestations
- The clinical manifestations depend on
 - Age of patient
 - Immune status
 - Previous immunity
 - Antigenic type of the virus
 - Anatomical site of involvement
- HSV-1 produces the lesions above the waist,
- HSV-2 produces lesions below the waist

Immunity

- New born-passive 6 months
- 6 mo-2 y-highly susceptible
- Transplacental antibodies↓ infection
- Primary infection →antibodies +virus↓

latency

Recurrent → antibodies change/modify subsequent disease

- Both antibody-mediated and cell-mediated reactions are clinically important.
- CMI plays an important role in conferring immunity to HSV infection and facilitates recovery

Diagnosis

- Light microscopy-intranuclear inclusions
infected cell → ballooning & fusion
- Electron microscopy
- Antigen detection
- Virus culture-HSV 1, HSV 2 easiest to culture typical CPE
- Serology
- DNA sequencing can be used to distinguish between the two types and among strains of each subtype.

Antiviral chemotherapy

Inhibit DNA synthesis-inhibit viral replication

- Acyclovir-drug of choice
- Famiclovir
- Valaciclovir better absorption
- Idoxuridine
- Vidarabine

HUMAN HERPES VIRUS TYPE-3

- VARICELLA- ZOSTER VIRUS

Two almost universal human diseases

- 1 Chickenpox (Varicella)-exanthema of childhood
- 2 Herpes zoster (Shingles)

Disabling disease of aged persons or immunocompromised patients

- a ubiquitous and extremely contagious infection
- has the smallest genome of all the human herpesviruses
- Transmitted mainly by the respiratory secretions
- Intranuclear inclusions
- CPE more focal much more slow
- Only one serotype X HSV

Varicella-Zoster

- Varicella-Chicken pox ↓

Latency ↓

Zoster-Shingles

two distinct clinical entities

- Both diseases same virus
- Morphologically identical HSV
- Only one serotype-x HSV
- No animal reservoir (except primates)
- Grow readily cell culture
- Intranuclear inclusions, ballooning, swelling

Varicella-Zoster Virus

Normal individuals

- Primary infection (chickenpox) is one of the **classical rash** diseases of childhood.
- Following primary infection, the virus remains latent in the **cranial-spinal ganglia**.
- Reactivation leading to the appearance of **shingles** occurs in 10-20% of infected individuals and usually occurs after the **fourth** decade of life. Usually, only one episode of reactivation occurs.

Properties of VZ virus

- a ubiquitous and extremely contagious infection
- has the smallest genome of all the human herpesviruses
- Transmitted mainly by the respiratory secretions
- Intranuclear inclusions
- CPE more focal much more slow
- Only one serotype X HSV

Diagnosis

- Clinical
- EM diff poxviruses
- Cytology-multinucleated giant cells
- Intracellular viral antigen- **confirmation**
- Serology
- Molecular methods-PCR,EIA

Immunity

- Primary Varicella-life long immunity to Varicella
- Zoster can occur
- CMI important in recovery

Cytomegalovirus(HHV 5)

- The largest of the Herpesviruses, genome ~240kbp
- Linear, ds-DNA virus and grows only in diploid human fibroblast cell culture
- CMV infection are 'slow' - 7-14 days;
- CMV infection is common more than 50 % population experienced infection by the age of 40
- Most infections are asymptomatic **except in people** with immune defects (**T-cell defects**) /pregnancy / newborns (congenital)

Cytomegalovirus

Normal individuals

- Primary infection is usually asymptomatic, occasionally an infectious **mononucleosis**-like illness may be seen.
- Reactivations or re-infections are common throughout life and are usually asymptomatic.

Immunocompromised individuals

- Both primary and recurrent infection may lead to symptomatic disease.
- Primary CMV infection is usually more severe than recurrent infection, with the **exception** of **bone marrow transplant** recipients, where primary and recurrent infections are just as severe.

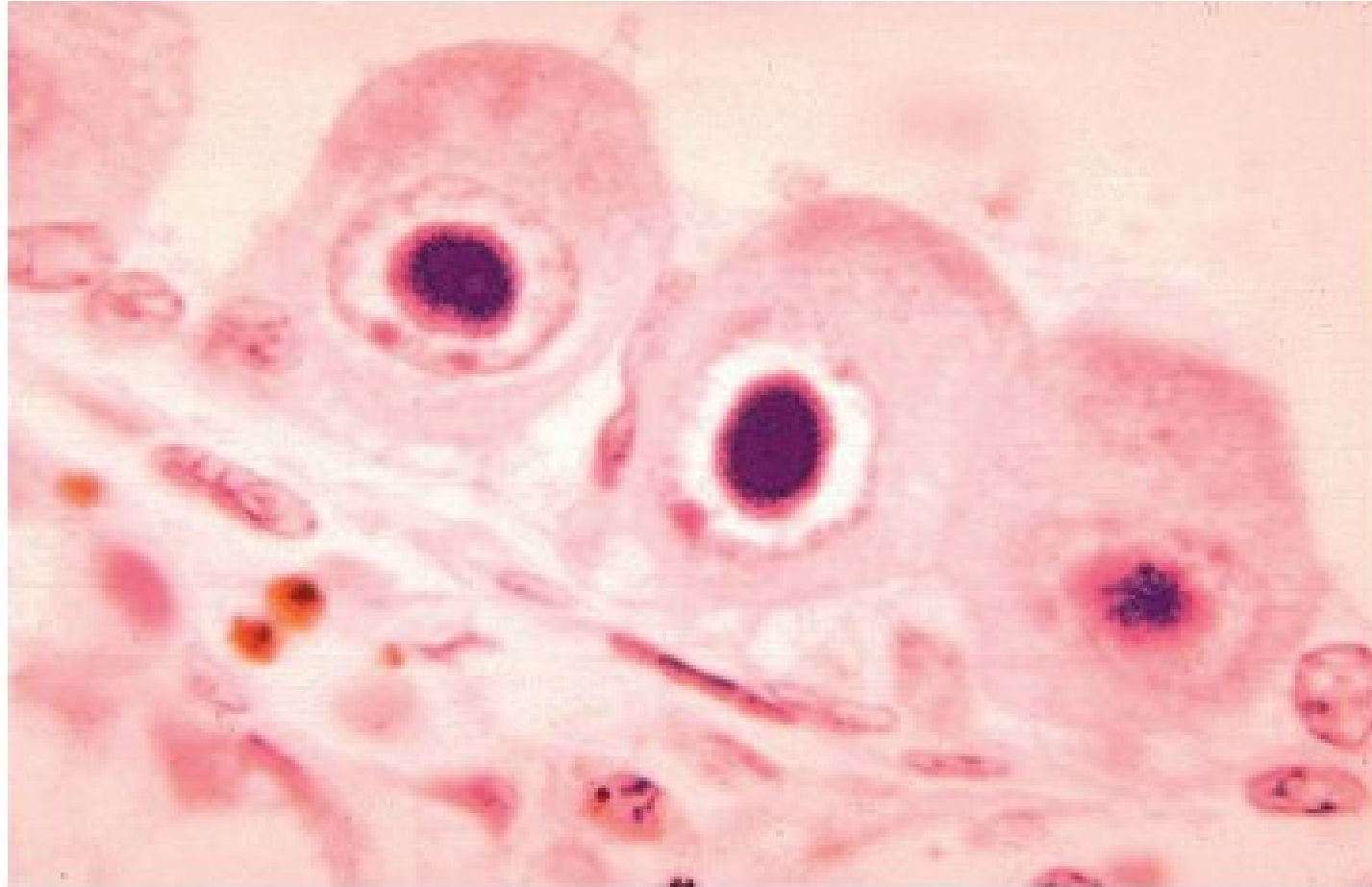
Clinical Manifestations

- Infection is acquired by coming in contact with blood, tissue, and most body secretions containing viruses
 - Fever
 - Hepatitis
 - Gastrointestinal manifestations eg. colitis
 - Encephalopathy
 - Retinitis
 - Poor graft function
 - Pneumonitis, the most severe manifestation, and carries a mortality rate of 85% in the absence of treatment.

Laboratory Diagnosis

- 1 Cytology / histology –large cytomegalic 25-35um intranuclear inclusions-”owl’s eye”
- 2 Culture -Gold standard 4-6 weeks
- 3 Antigen detection
- 3 Serology
- 3 PCR

Cytopathic Effect of CMV



Epstein-Barr Virus(HHV 4)

- EBV is the ultimate B-lymphocyte parasite
 - Acute infectious mononucleosis / nasopharyngeal carcinoma
 - Burkitt's Lymphoma and other lymphoproliferative disorders
 - Dual cell tropism for human B-lymphocytes (generally non-productive infection) and epithelial cells (productive infection).
 - Highly host specific-no suitable animal host
- ❑ Two viral types: EBV1 and EBV2 depending on:
 - variation in genome, Structure, nuclear antigen and biologic properties

Clinical manifestation

- The diseases of EBV result from either
 - overactive immune response (infectious mononucleosis)
 - lack of effective immune control (lympho-proliferative disease and hairy cell leukoplakia)
- **Laboratory Diagnosis:** EBV-induced infectious mononucleosis is based on the three classic criteria
 - Presence of lymphocytosis
 - Presence of at least 10% atypical lymphocytes in peripheral blood smear
 - Presence of heterophilic antibodies and antibodies to viral antigens
- These criteria are supplemented by direct detection of viral antigens or EBV genomes in clinical specimens

Epstein-Barr Virus-clinical---

Young adults-

- **Infectious mononucleosis**-autoantibodies

Self limiting lasts 2-4 weeks

Symptoms like primary infection

- **Oral Hairy Leukoplakia**: Wart like growth on tongue of some HIV persons and transplant patients. It is an epithelial focus of EBV replication
- **Burkitt's lymphoma**

Burkitt's lymphoma

- Tumor of jaw
- African children/ young adults

Contain EBV DNA

Most cases express EBNA 1 antigen

HHV-6 and HHV-7

- HHV-6 and HHV-7
- Develop latency following primary infection in T cells and monocytes
- Reactivated from time to time in immune suppressed individual
- HHV-6 infection is firmly associated with roseola infantum (exanthem subitum).
- It is associated with :
 - Neurological manifestations(febrile convulsions, meningitis, and encephalitis)
 - A variety of symptoms in transplant recipients such as fever, graft vs host disease, liver and CNS manifestations.
- HHV-7 is not associated conclusively with any human disease.

Human herpes Virus 6 and 7

- HHV-6 and HHV-7 become latent following primary infection → Reactivation
- T- lymphotropic human viruses
- HHV 6-1986,
- grows well in CD4
- Other cells like B cells too support growth
- Oropharynx → saliva-transmission
- HHV-6-Early life-roseola infantum “sixth disease” → persists for life – reactivation in pregnancy-encephalitis? neurological manifestations

Human Herpes Virus 8

- associated with **Kaposi's sarcoma**
- as well as some lesser known malignancies such as **Castleman's disease** and primary effusion lymphomas.
- the B cell is the primary target cell
- HHV-8 DNA is found in almost 100% of cases of Kaposi's sarcoma.
- Most patients with KS have antibodies against HHV-8.
- HHV-8 **does not have a ubiquitous distribution**
- Homosexuals
- Fosarnet, Ganiclovir, Cidofovir.

Human Parvo viruses

- Parvoviruses are the smallest viruses and simplest DNA animal viruses
- Posses ssDNA genome
- One known human pathogen (parvovirus B19)
- Dependent viruses, replicating host cell / coinfecting helper viruses
- Virion-Icosahedral, DNA 20 %, protein 80 %
- Proteins: One major (VP2) and one minor (VP1)
- **No envelope**
- Replication Nucleus, dependent on dividing host cell (S-phase)

- Smallest of all human viruses (18-26 nm)
- Human pathogen only- B19-no other host
- Tropism erythroid (RBC) progenitors



Erythema infectiosum “fifth disease”

- Some Parvoviruses cause serious problems in cats and dogs
- The virus do not contain lipids or carbohydrates
 - Inactivated by formalin, β -propiolactone, oxidizing agents & γ -irradiation
 - Stable over a wide range of pH (3-9)
 - Resistant to lipid solvents

Classification

- Family Parvoviridae
- 2 Subfamilies
 - ❖ 1 Parvovirinae – Vertebrates
 - 3 genera
 - 1. Parvoviruses-Feline panleukopenia virus
 - 2. Erythrovirus – B19 **only one serotype**
 - 3. Dependovirus- adenovirus associated virus
 - ❖ 2. Densovirinae - Insects .

PATHOGENESIS AND IMMUNITY

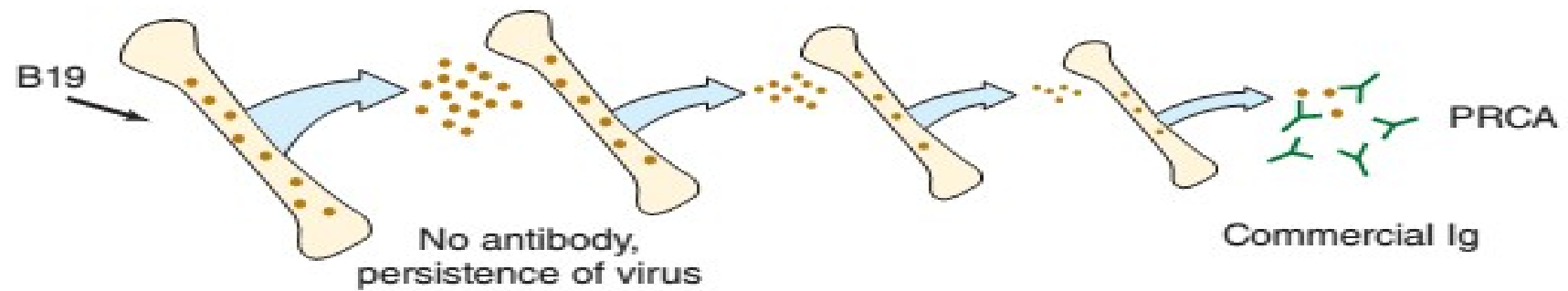
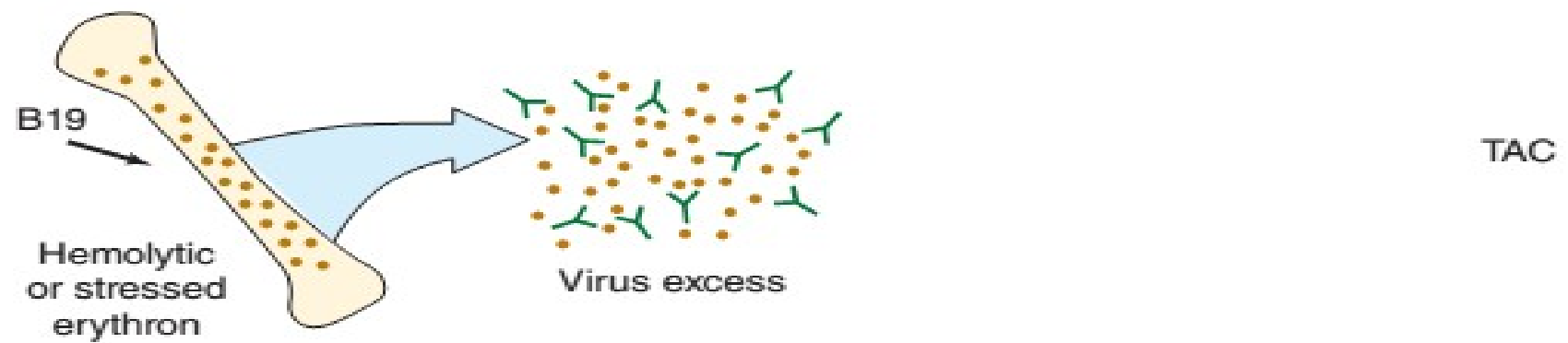
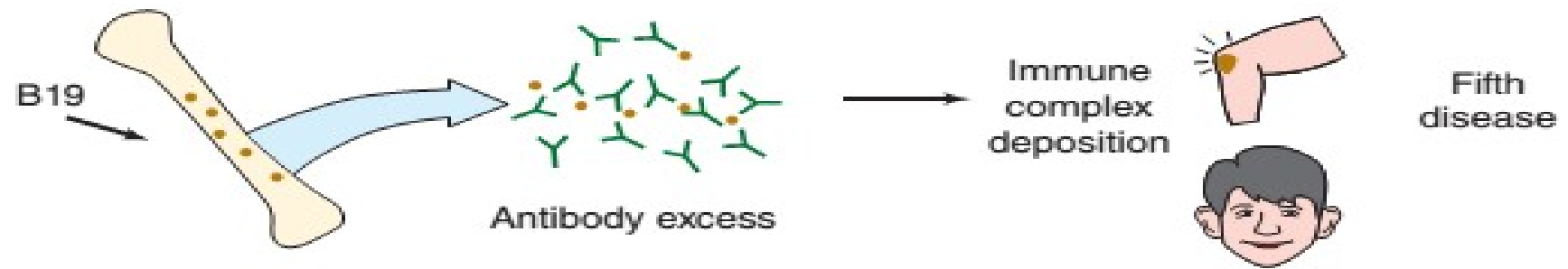
- B19 targets and is cytolytic for erythroid precursor cells
- Major sites of virus replication in patients are assumed to be the **adult marrow**, **some blood cells**, and the **fetal liver**
- In immunocompromised patients, persistent B19 infections occur, resulting in chronic anemia.
- B19 virus first replicates in the **nasopharynx** or upper respiratory tract then spreads by viremia to the **bone marrow**
- B19 viral disease has a **biphasic course**

- Initial febrile stage erythrocyte production is stopped for 1 week
 - *the infectious stage*
 - A large viremia occurs within 8 days of infection
 - Large numbers of virus are also released into oral and respiratory secretions.
 - Antibody stops the viremia and is important for resolution
- The second, symptomatic, stage is immune mediated.
 - rash and arthralgia seen
 - virus-specific antibody
 - disappearance of detectable B19 virus,
 - formation of immune complexes

PATHOGENESIS of B19

- B19 infection → antibody excess → Immune complex deposition → Fifth disease
- B19 infection → Virus excess → Transient Aplastic Crisis (TAC)
- B19 infection → No antibody, persistence of virus → PRCA pure red cell aplasia
- Fetal infection → fetal heart, bone marrow, liver affected → Hydrops fetalis

A



Diagnosis

Specimens:

- Serum (principal specimen)
- Tissue biopsy

A. Virus Detection

Culture of B 19 in erythroid progenitor cells derived from:

- Human bone marrow, umbilical cord, peripheral blood, or fetal liver sources
- Failed to grow in conventional cell culture lines & animal model

B. Serologic tests

- ELISA (detection of B19-specific IgM & IgG antibodies)
- B19 IgM recent infection, IgG antibody persist for years
- Haemagglutination-based assays

C. Molecular technique

- Detection of **viral DNA** by quantitative PCR is the mainstay of detection of B19V
- Low levels of viral DNA (**$<10^4$ IU (genome copies) ml⁻¹**)

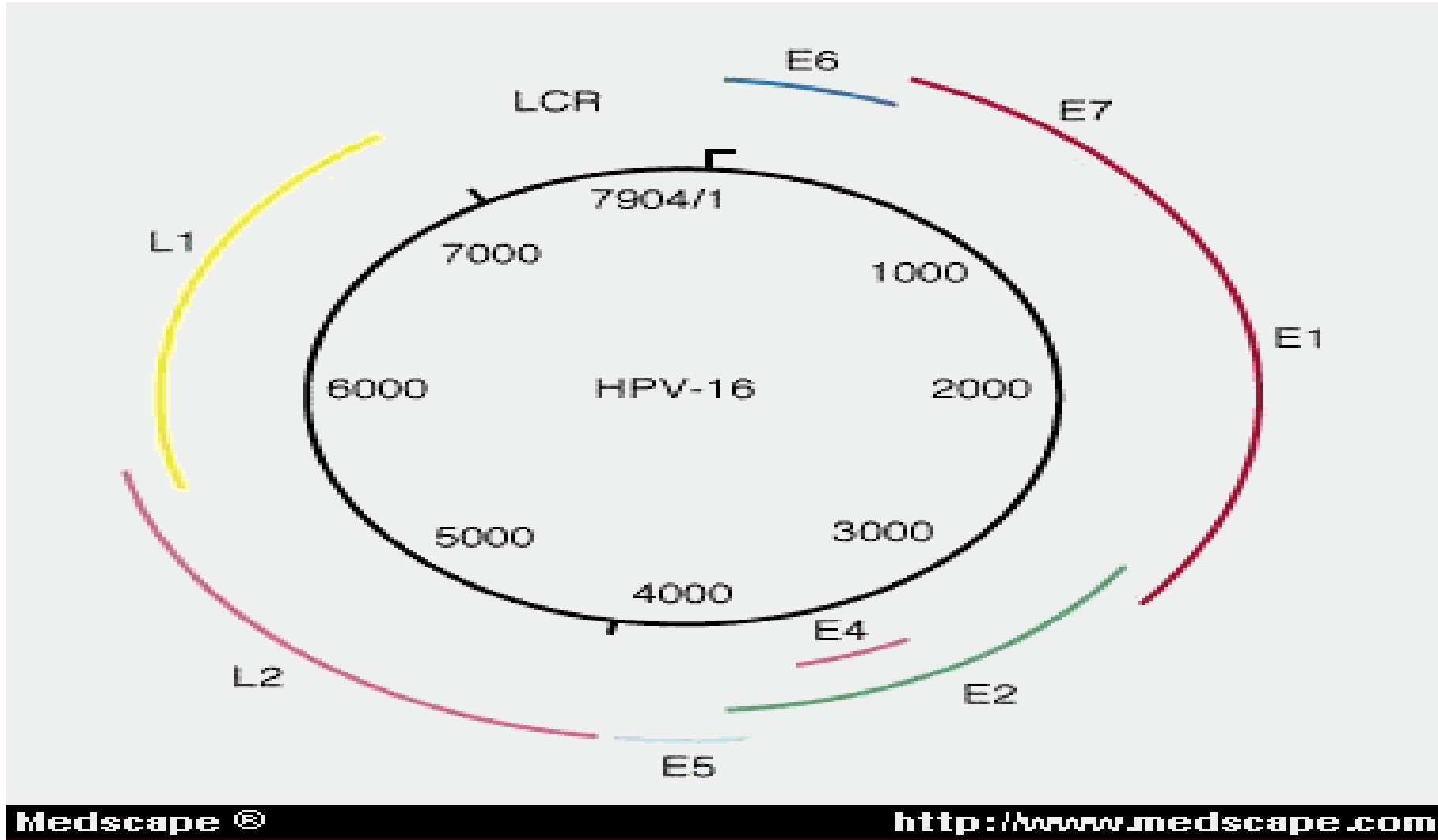
can be detected for months, or even years after acute infection

Human Papillomaviruses (HPV)

Properties

- Genome is circular dsDNA
 - Encodes seven to eight early genes (E1–E8) and two late (L1 and L2) genes.
 - All these genes are present in a single strand, that is the “ + ” strand.
 - E6 and E7 are implicated in carcinogenesis
 - E1 and E2 participate in DNA replication
- More than 100 types classified in to 16 (A-P) groups associated with **cervical cancer**
- Non-enveloped with icosahedral symmetry
- Major & minor capsid protein comprises outer protein coat of the virus

- Three major regions comprise the HPV genome:
 - **Early region** consists of genes responsible for **transcription**, plasmid **replication**, & **transformation**
 - The **late region** codes for the major (L1) and minor (L2) capsid proteins encode for expression of structural proteins
 - **Control region** contain the regulatory elements for transcription and replication
- Shows tropism for epithelial cells of the skin and mucous membranes
- Undergo cell transformation



HPV gene products & their function

Gene	Function
E1	Initiation of DNA replication
E2	Transcriptional regulation/DNA replication
E3	?
E4	Disrupts cytoskeleton?
E5	Transforming protein, interacts with growth factor receptors
E6	Transforming protein, binds to p53 gene, leading to degradation
E7	Transforming protein, binds to pRB
E8	?
L1	Major capsid protein
L2	Minor capsid protein

Human Papillomaviruses

- Several different kinds of warts in humans
 - Skin warts
 - Plantar warts
 - Flat warts
 - Genital condylomas
 - Laryngeal papillomas

Transmission

1. Sexual contact

- Greater than 95% of infection is through sexual contact
- In children associated with sexual abuse

2. Vertical transmission

- Less frequent mode of transmission
- Difficult to detect due to the latency period between the infant's exposure at birth & symptom presentation

3. Other pathways

- Non-sexual transmission, e.g. contact with infected urogenital secretions or bathing together
 - ▶ No such cases have been clearly documented???

Risk factors for HPV

1. Sexual behavior

- Is a primary risk factor for infection
- Women with multiple sex partners have a higher risk than monogamous women

2. Immune suppression

- Is a risk factor for all people exposed to HPV
- A person with a pre-existing immuno-compromised state and/or concurrent genital infection has a 17-fold increased risk of developing the diseases

3. Age

- Young women, between the ages of 15 and 25 have a two fold higher risk of developing an HPV infection than women over 35

4. Other possible risk factors

Pregnancy, smoking , concurrent herpes infections , others

- ▶ Associated with increased HPV infection in several studies, but their significance is not conclusive

5. Socioeconomic variables

Poverty, domestic violence, sexual abuse, inadequate health Care & lack of information

- ▶ Can facilitate disease transmission , prevent early detection & treatment

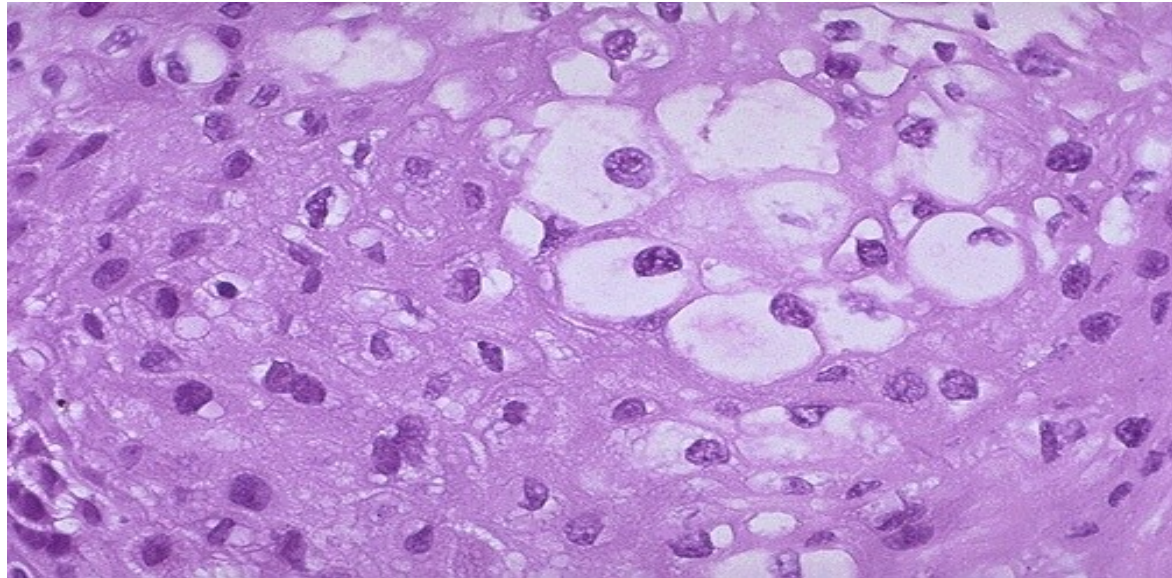
HPV Vaccine

- vaccine tetravalent ([Gardasil](#): HPV 6,11,16,18) or divalent ([Cervarix](#): HPV-16 and -18) HPV vaccine is recommended
- HPV girls and women between 9-26 years
- Efficacy of vaccine ~ 100 % if given before sexual exposure
- US recommendations- Routine HPV vaccine 13-26 age females (can be initiated at 9 years)

Diagnosis

- Pap smears must be done carefully and timely
- Wart- confirmed microscopically histologic appearance
- DNA molecular probes
- PCR
- Virus does not grow in cell culture
- Tests for HPV antibodies rarely

Pap stained cervical smear



An abnormal Pap Smear showing changes consistent with Human Papilloma Virus (HPV) and Mild Dysplasia. Note the irregular **peri-nuclear cytoplasmic clearing**, which is the key characteristic of identifying HPV visually on a Pap Smear.



Viral Hepatitis

- Viral hepatitis is a systemic disease primarily involving the liver.
- Most cases of acute viral hepatitis in children and adults are caused:
 - Hepatitis A virus (HAV) (infectious hepatitis)
 - Hepatitis B virus (HBV) (serum hepatitis)
 - Hepatitis C virus (HCV) (common cause of post-transfusion hepatitis)
 - Hepatitis D (HDV), a defective virus
 - Hepatitis E virus (HEV) enterically transmitted hepatitis
- Hepatitis viruses produce acute inflammation of the liver

Clinical picture

- Acute hepatitis- Jaundice, epigastric discomfort, nausea., and fever
- Chronic hepatitis- fatigue, exhaustion, low vitality, GI symptoms
- Advanced cases- Jaundice, liver cirrhosis, hepatomegaly

Hepatitis A-G Overview

	HAV	HBV	HCV	HDV	HEV	HGV
Family	Picorna- viridae	Hepadna- viridae	Flavi- viridae	Viroid (?)	Calici- viridae	Flavi- viridae
Structure						
Virion						
envelope	no	yes	yes	yes	no	yes
genome	(+) RNA	ds DNA	(+) RNA	(-) RNA	(+) RNA	(+) RNA
Transmission	feacal-oral	parenteral	parenteral	parenteral	feacal -oral	parenteral
Incubation period	15-49d	25-160d	21-84d	60-110d	10-56d	?
Chronic disease	no	5-90%	80%	10-90%	no	yes
HCC	no	yes	yes	yes	no	?
Therapy	no	yes	yes	yes	no	no
Vaccination	yes	yes	no	yes	no	no

HEPATITIS A VIRUS

- Hepatitis A Virus: Structure and Classification
 - RNA Picornavirus
 - Seven genotypes
 - Heat and acid stable
 - Always acute disease - never chronic-not oncogenic
- **Highly resistance:** stable 20 % ether, acid (pH 1.0 for 2 hrs); 60 C -1 hour; Drying - 1 month
- **Destroyed:** Autoclaving, Boiling-5 mts, Dry heat 180 C-1 hr, UV irradiation, Chlorine, Heating food >85 for 1 mt and Formalin

Hepatitis A Diagnosis

- Detection of IgM antibody-active disease **ELISA**
- IgG antibodies soon after onset-persist for decades - suggests prior infection or vaccination.
 - appears to provide lifelong immunity against recurrent infection
- The classic findings in the hepatocytes include
 - Mononuclear infiltrate,
 - Ballooning,
 - Degeneration, and
 - Acidophilic (*councilman-like*) bodies

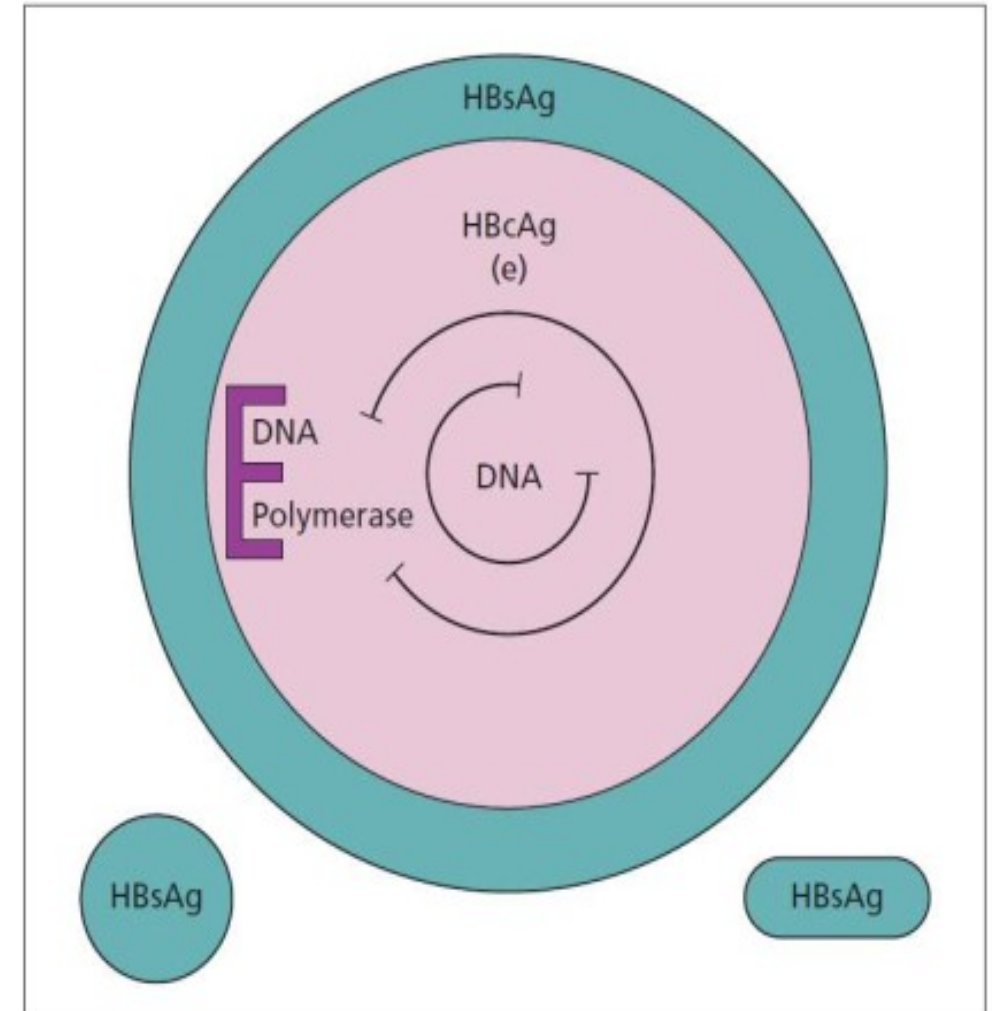
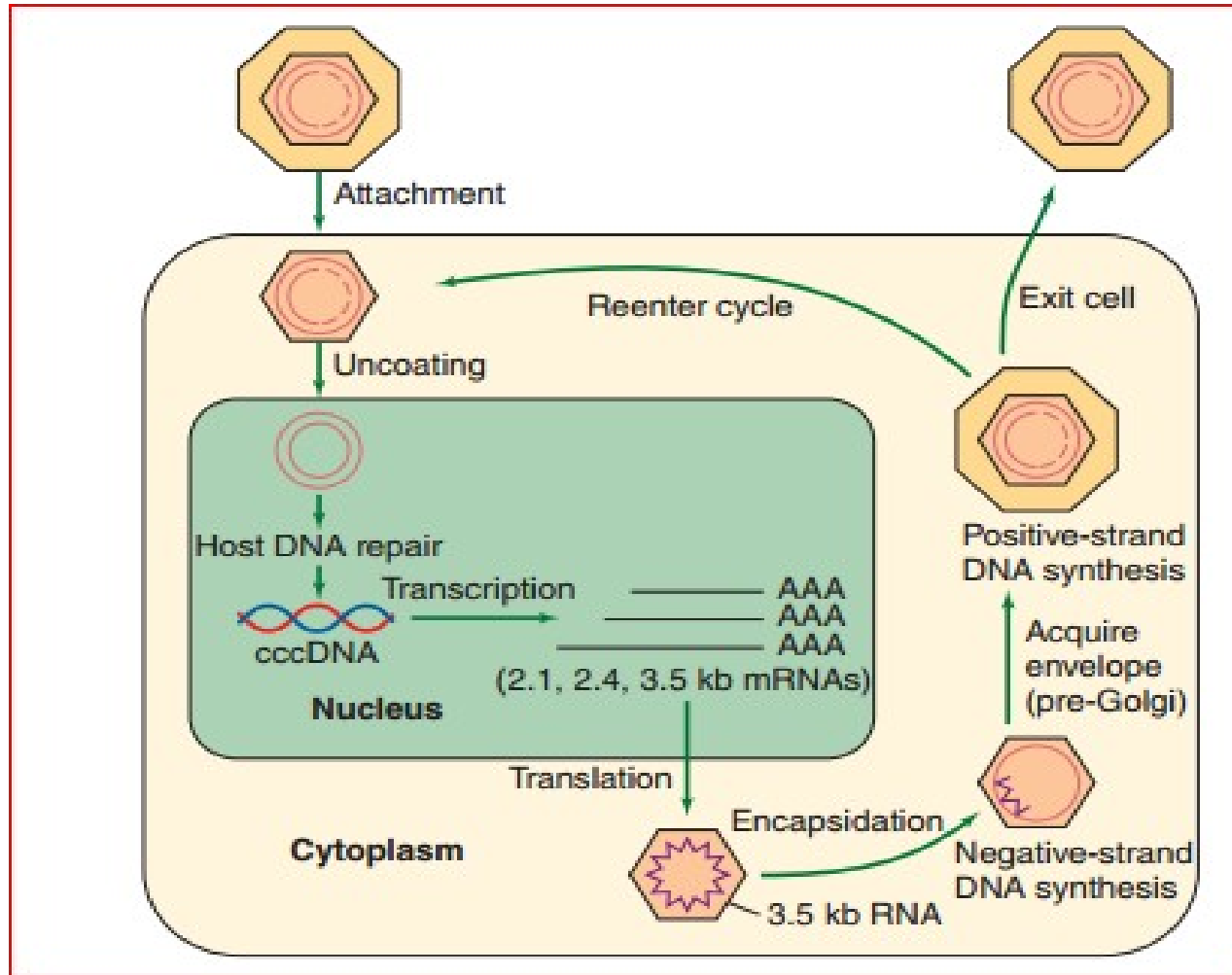
Hepatitis B Virus

- Family- Hepadnaviridae, genus-Orthohepatadnavirus
- Virion also referred to as **Dane** particle
- Enveloped virion with a genome of a small, circular, partially double-stranded DNA
- Replication occurs through an RNA intermediate, reverse transcriptase
- Core antigens located in the center (nucleocapsid)
 - Core antigen (HBcAg)
 - e antigen (HBeAg)- an indicator of transmissibility (minor component of the core- antigenically distinct from HBcAg)

- The complete negative strand possesses four genes; S, C, P, and X
 - S codes for HBsAg and also for HBeAg (hepatitis B e antigen)
 - P codes for DNA polymerase that has **reverse transcriptase** activity
 - Gene X codes for the X protein that has **transcription-regulating** activity.
- The HBV consists of nucleocapsid which surrounds HBV DNA and DNA polymerase with reverse transcriptase and ribonuclease activity

- Have limited tissue tropisms and host ranges, Infects the **liver** and, to a lesser extent, the **kidneys** and **pancreas**
- They resist treatment with ether, low pH, freezing, and moderate heating.
- HBV express three main antigens:
 - HBsAg: It indicates current hepatitis B infection, marker of infectivity .
 - HBcAg: marks the replication of the virus
 - HBeAg
 - These antigens are recognized by antibody proteins that bind specifically to one of these surface proteins
 - HBsAg, originally termed the *Australia antigen*, includes three glycoproteins (L, M, and S)
- **window period**

Replication



cccDNA: covalently closed circular DNA

Clinical outcomes of Hepatitis B infections

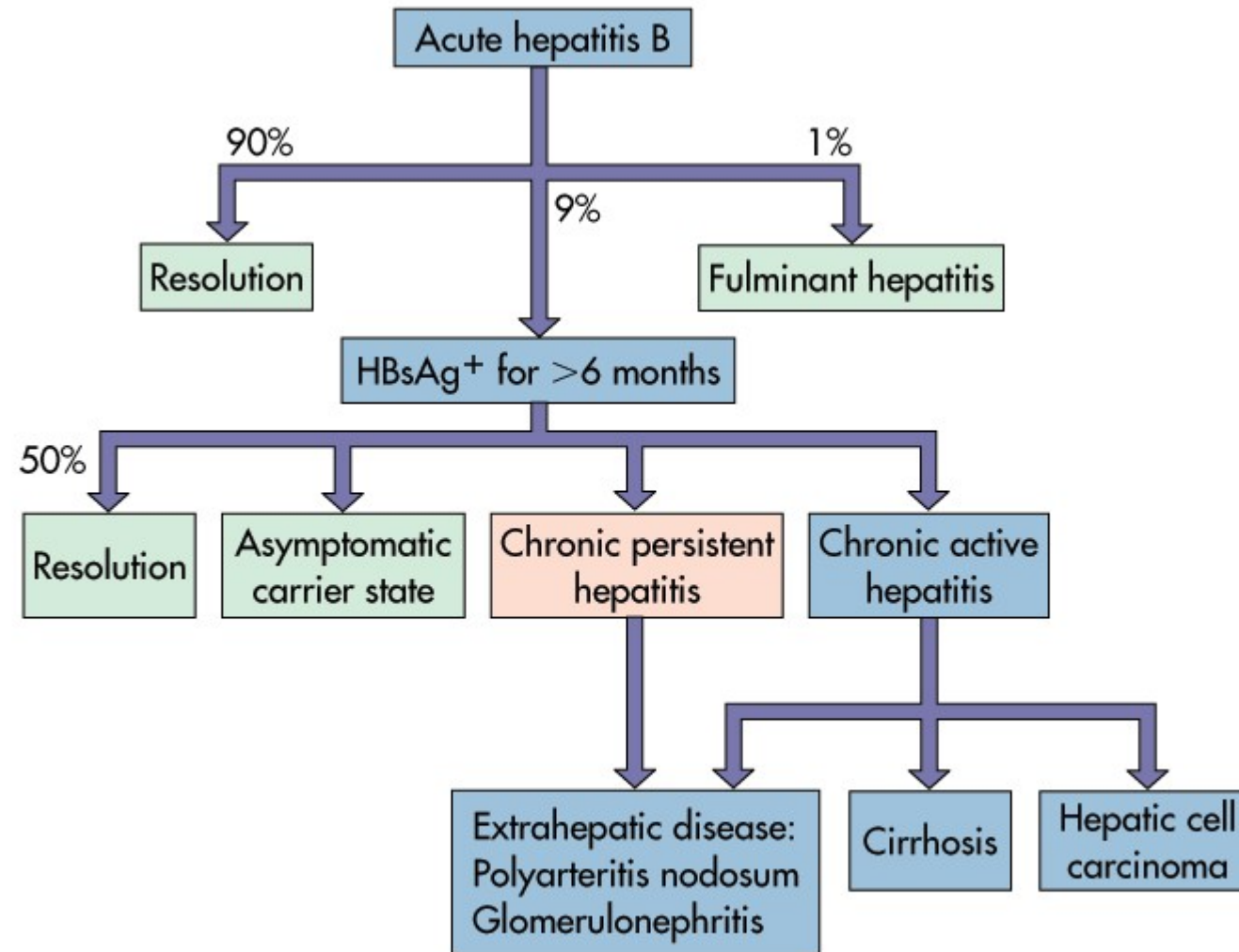


Figure 66-11. Clinical outcomes of acute hepatitis B infection. (Redrawn from White DO, Fenner F: *Medical virology*, ed 3, New York, 1986, Academic Press)

Modes of transmission of Hepatitis B

- **Parenteral route---** Blood transfusion | Blood handling | accidental inoculation | immunization | tattooing | circumcision
- **Sexual transmission---** (50-100 times easier to transmitted than HIV)
- **Perinatal transmission---** Mothers with HBeAg positive are much more likely to transmit to their offspring in high prevalence populations
- **Occupational exposure---** (needle stick injuries) transmission rate 30- 60%)
- **Concentration of hepatitis B virus in various body fluids**
 - **High:** blood, serum and wound exudates
 - **Moderate:** semen vaginal and saliva
 - **Low/Not Detectable:** urine, feces, sweat, tears and breastmilk

High-risk groups for HBV infection

- **People from endemic regions**
- **Babies of mothers with chronic HBV**
- **Intravenous drug abusers**
- **People with multiple sex partners**

Diagnosis

- **HBsAg** - used as a general marker of infection.
- **HBsAb** - used to document recovery and/or immunity to HBV infection.
- **anti-HBc IgM** - marker of acute infection.
- **anti-HBcIgG** - past or chronic infection.
- **HBeAg** - indicates active replication of virus and therefore infectiveness.
- **Anti-Hbe** - virus no longer replicating. However, the patient can still be positive for HBsAg which is made by integrated HBV.

Diagnosis

A battery of serological tests are used for the diagnosis of acute and chronic hepatitis B infection.

Table 18.5. Interpretation of HBV serological markers

HBsAg	HBV infection: acute or chronic
HBeAg	High levels of HBV replication and infectivity
Anti-HBe	Low levels of HBV replication and infectivity
Anti-HBc (IgM)	Recent HBV infection
Anti-HBc (IgG)	Recovered or chronic HBV infection
Anti-HBs	Immunity to HBV infection
Anti-HBc (IgG) + anti-HBs	Past HBV infection
Anti-HBc (IgG) + HBsAg	Chronic HBV infection

Hepatitis C Virus

- Member of the Flaviviridae family (other members yellow fever and dengue)
- Enveloped single stranded RNA virus
- It has a very simple genome with three structural and five nonstructural genes
- The major agents to cause **NANB** hepatitis
- Antigens of hepatitis C are not detectable in blood, so diagnostic tests attempt to demonstrate antibody.
- Infection with the hepatitis C virus (HCV) remains chronic in 70-85% of infected individuals
- In 20-40% cases of chronic hepatitis C
 - leads to end-stage liver diseases: cirrhoses, hepatocellular carcinoma and liver failure after 20-30 years of HCV infection

Diagnosis

- **HCV antibody** by serology
- **HCV-RNA** –
 - **PCR: used to diagnose HCV infection in the acute phase**
: its main use is in monitoring the response to antiviral therapy

Quiz 1.

1. List the basic structural component of a virus
2. Give at least three mechanisms used for the classification of viruses
3. What are the basic steps of virus replication
4. What is the basis of Baltimore classification of medically important viruses
5. Which of the agents of Viral hepatitis replicate through RNA intermediate